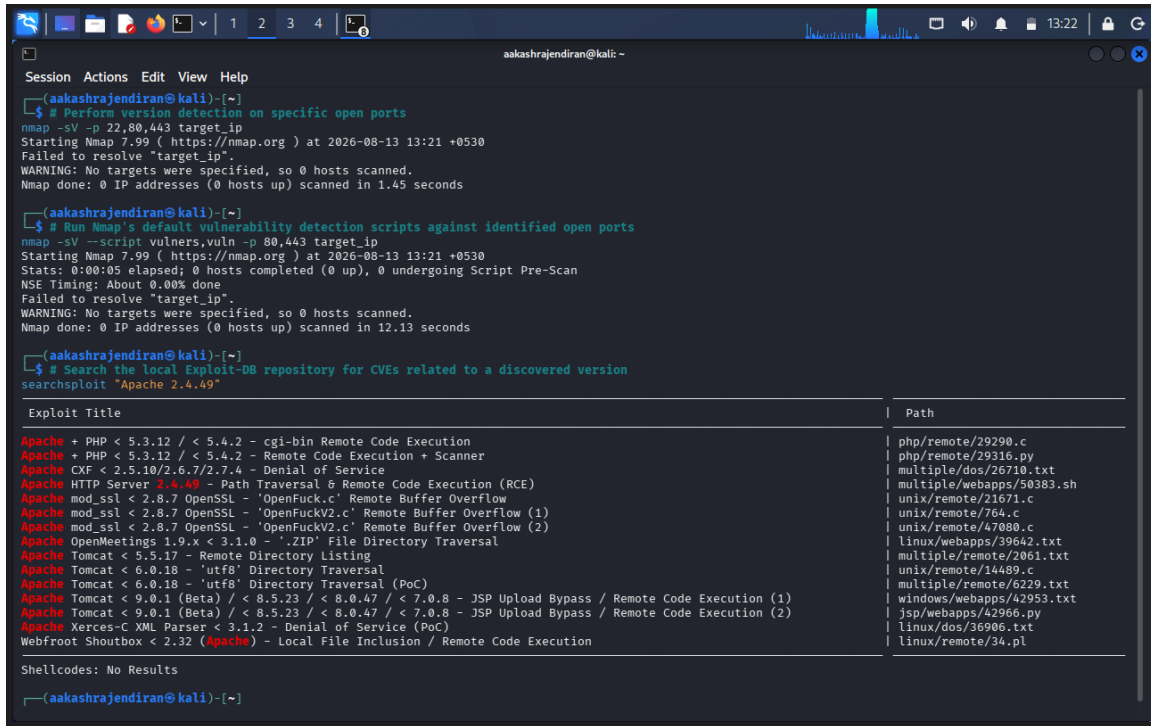


Q1.

```
sudo nmap -sV -p 22,80,443 target_ip  
sudo nmap -sV --script vulners,vuln -p 80,443 target_ip  
searchsploit "Apache 2.4.49"
```



```
aakashrajendiran@kali: ~  
Session Actions Edit View Help  
aakashrajendiran@kali)~  
$ # Perform version detection on specific open ports  
nmap -sV -p 22,80,443 target_ip  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:21 +0530  
Failed to resolve "target_ip"  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.45 seconds  
aakashrajendiran@kali)~  
$ # Run Nmap's default vulnerability detection scripts against identified open ports  
nmap -sV --script vulners,vuln -p 80,443 target_ip  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:21 +0530  
Stats: 0:00:05 elapsed; 0 hosts completed (0 up), 0 undergoing Script Pre-Scan  
NSE Timing: About 0.00% done  
Failed to resolve "target_ip"  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 12.13 seconds  
aakashrajendiran@kali)~  
$ # Search the local Exploit-DB repository for CVEs related to a discovered version  
searchsploit "Apache 2.4.49"  


| Exploit Title                                                                                                | Path                      |
|--------------------------------------------------------------------------------------------------------------|---------------------------|
| Apache + PHP < 5.3.12 / < 5.4.2 - cgi-bin Remote Code Execution                                              | php/remote/29290.c        |
| Apache + PHP < 5.3.12 / < 5.4.2 - Remote Code Execution + Scanner                                            | php/remote/29316.py       |
| Apache CXF < 2.5.10/2.6.7/2.7.4 - Denial of Service                                                          | multiple/dos/26710.txt    |
| Apache HTTP Server 2.4.49 - Path Traversal & Remote Code Execution (RCE)                                     | multiple/webapps/50383.sh |
| Apache mod_ssl < 2.8.7 OpenSSL - 'OpenFuck.c' Remote Buffer Overflow                                         | unix/remote/21671.c       |
| Apache mod_ssl < 2.8.7 OpenSSL - 'OpenFuckV2.c' Remote Buffer Overflow (1)                                   | unix/remote/764.c         |
| Apache mod_ssl < 2.8.7 OpenSSL - 'OpenFuckV2.c' Remote Buffer Overflow (2)                                   | unix/remote/47080.c       |
| Apache OpenMeetings 1.9.x < 3.1.0 - '.ZIP' File Directory Traversal                                          | linux/webapps/39642.txt   |
| Apache Tomcat < 5.5.17 - Remote Directory Listing                                                            | multiple/remote/2061.txt  |
| Apache Tomcat < 6.0.18 - 'utf8' Directory Traversal                                                          | unix/remote/14489.c       |
| Apache Tomcat < 6.0.18 - 'utf8' Directory Traversal (PoC)                                                    | multiple/remote/6229.txt  |
| Apache Tomcat < 9.0.1 (Beta) / < 8.5.23 / < 8.0.47 / < 7.0.8 - JSP Upload Bypass / Remote Code Execution (1) | windows/webapps/42953.txt |
| Apache Tomcat < 9.0.1 (Beta) / < 8.5.23 / < 8.0.47 / < 7.0.8 - JSP Upload Bypass / Remote Code Execution (2) | jsp/webapps/42966.py      |
| Apache Xerces-C XML Parser < 3.1.2 - Denial of Service (PoC)                                                 | linux/dos/36906.txt       |
| Webfoot Shoutbox < 2.32 (Apache) - Local File Inclusion / Remote Code Execution                              | linux/remote/34.pl        |

  
Shellcodes: No Results  
aakashrajendiran@kali)~
```

Q2.

```
sudo nmap -sS -p 80,443 --packet-trace target_ip  
nmap -sT -p 80,443 --packet-trace target_ip  
sudo tcpdump -i eth0 host target_ip and port 80 -nn -v  
sudo hping3 -S -p 80 -c 3 target_ip  
nc -zv -w 3 target_ip 80
```

```
Session Actions Edit View Help
(aakashrajendiran@kali)-[~]
$ # Determine if ports are filtered by firewall rules
sudo nmap -sA -p 21,22,80,443,3389 target_ip
[sudo] password for aakashrajendiran:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:26 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.21 seconds

(aakashrajendiran@kali)-[~]
$ # Send fragmented IP packets (-f)
sudo nmap -sS -f -p 80,443 target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:27 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 5.96 seconds

(aakashrajendiran@kali)-[~]
$ # Spoof source port 53 (DNS)
sudo nmap -sS --source-port 53 -p 80,443 target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:27 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.63 seconds

(aakashrajendiran@kali)-[~]
$ # NULL Scan (No flags set)
sudo nmap -sN -p 80,443 target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:27 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.38 seconds

# FIN Scan (FIN flag set)
sudo nmap -sF -p 80,443 target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:27 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.46 seconds

# Xmas Scan (FIN, PSF, URG flags set)
sudo nmap -sX -p 80,443 target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:27 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.46 seconds
```

Q3.

```
sudo nmap -sA -p 21,22,80,443,3389 target_ip

sudo nmap -sS -f -p 80,443 target_ip

sudo nmap -sS --source-port 53 -p 80,443 target_ip

sudo nmap -sN -p 80,443 target_ip

sudo nmap -sF -p 80,443 target_ip

sudo nmap -sX -p 80,443 target_ip

sudo nmap -sS -D 192.168.1.5,192.168.1.6,ME -p 80,443 target_ip
```

```
Session Actions Edit View Help
(aakashrajendiran@kali)-[~]
$ # Perform a fast, half-open TCP SYN scan (Requires sudo)
sudo nmap -sS -p 1-1000 target_ip
[sudo] password for aakashrajendiran:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:29 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.01 seconds

(aakashrajendiran@kali)-[~]
$ # Perform a full-connect scan without root/sudo privileges
nmap -sT -p 22,80,443 target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:29 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 5.74 seconds

(aakashrajendiran@kali)-[~]
$ # Scan the top 20 most common UDP ports
sudo nmap -sU --top-ports 20 target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:29 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.21 seconds

(aakashrajendiran@kali)-[~]
$ # Service version detection on specific UDP ports (improves accuracy)
sudo nmap -sU -sV -p 53,67,123,161 target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:29 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.56 seconds

(aakashrajendiran@kali)-[~]
$
```

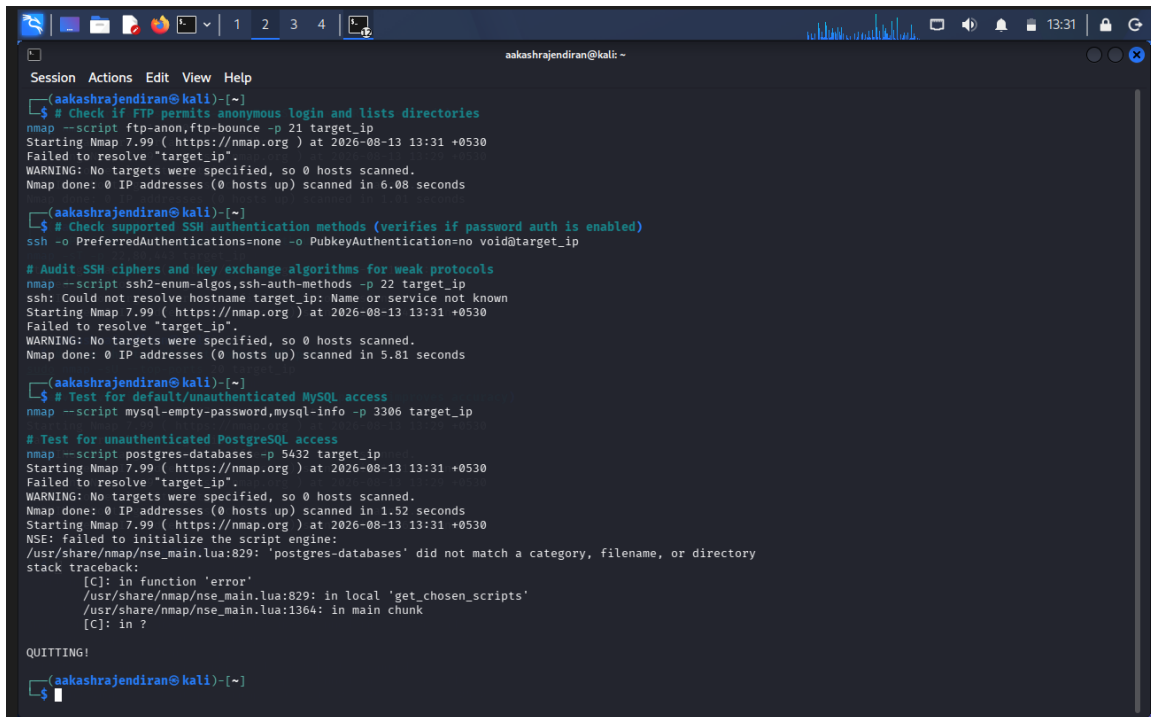
Q4.

```
sudo nmap -sS -p 1-1000 target_ip
```

```
nmap -sT -p 22,80,443 target_ip
```

```
sudo nmap -sU --top-ports 20 target_ip
```

```
sudo nmap -sU -sV -p 53,67,123,161 target_ip
```



```
aakashrajendiran@kali: ~  
Session Actions Edit View Help  
aakashrajendiran@kali)~$ # Check if FTP permits anonymous login and lists directories  
nmap --script ftp-anon,ftp-bounce -p 21 target_ip  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:31 +0530  
Failed to resolve "target_ip".  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 6.08 seconds  
  
aakashrajendiran@kali)~$ # Check supported SSH authentication methods (verifies if password auth is enabled)  
ssh -o PreferredAuthentications=none -o PubkeyAuthentication=no void@target_ip  
  
# Audit SSH ciphers and key exchange algorithms for weak protocols  
nmap --script ssh2-enum-algos,ssh-auth-methods -p 22 target_ip  
ssh: Could not resolve hostname target_ip: Name or service not known  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:31 +0530  
Failed to resolve "target_ip".  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 5.81 seconds  
  
aakashrajendiran@kali)~$ # Test for default/unauthenticated MySQL access  
nmap --script mysql-empty-password,mysql-info -p 3306 target_ip  
  
# Test for unauthenticated PostgreSQL access  
nmap --script postgres-databases -p 5432 target_ip  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:31 +0530  
Failed to resolve "target_ip".  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 1.52 seconds  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:31 +0530  
NSE: failed to initialize the script engine:  
/usr/share/nmap/nse_main.lua:829: 'postgres-databases' did not match a category, filename, or directory  
stack traceback:  
  [C]: in function 'error'  
  /usr/share/nmap/nse_main.lua:829: in local 'get_chosen_scripts'  
  /usr/share/nmap/nse_main.lua:1364: in main chunk  
  [C]: in ?  
  
QUITTING!  
aakashrajendiran@kali)~$
```

Q5.

```
nmap --script ftp-anon,ftp-bounce -p 21 target_ip  
  
ssh -o PreferredAuthentications=none -o PubkeyAuthentication=no void@target_ip  
  
nmap --script ssh2-enum-algos,ssh-auth-methods -p 22 target_ip  
  
nmap --script mysql-empty-password,mysql-info -p 3306 target_ip  
  
nmap --script postgres-databases -p 5432 target_ip
```

```
Applications
Session Actions Edit View Help
aakashrajendiran@kali: ~
[~]$ # Stealth/SYN scan with packet trace
sudo nmap -sS -p 80,443 --packet-trace target_ip

# TCP Connect Scan with packet trace
nmap -sT -p 80,443 --packet-trace target_ip
[sudo] password for aakashrajendiran:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:24 +0530
NSOCK INFO [0.0500s] nsock_ioc_new2(): nsock_ioc_new (IOD #1)
NSOCK INFO [0.0500s] nsock_connect_udp(): UDP connection requested to 172.21.100.133:53 (IOD #1) EID 8
NSOCK INFO [0.0500s] nsock_ioc_new2(): nsock_ioc_new (IOD #2)
NSOCK INFO [0.0510s] nsock_connect_udp(): UDP connection requested to 172.21.100.31:53 (IOD #2) EID 16
NSOCK INFO [0.0510s] nsock_ioc_new2(): nsock_ioc_new (IOD #3)
NSOCK INFO [0.0510s] nsock_connect_udp(): UDP connection requested to 8.8.8.8:53 (IOD #3) EID 24
NSOCK INFO [0.0510s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 8 [172.21.100.133:53]
NSOCK INFO [0.0510s] nsock_read(): Read request from IOD #1 [172.21.100.133:53] (timeout: -1ms) EID 34
NSOCK INFO [0.0510s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 16 [172.21.100.31:53]
NSOCK INFO [0.0510s] nsock_read(): Read request from IOD #2 [172.21.100.31:53] (timeout: -1ms) EID 42
NSOCK INFO [0.0510s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 24 [8.8.8.8:53]
NSOCK INFO [0.0510s] nsock_read(): Read request from IOD #3 [8.8.8.8:53] (timeout: -1ms) EID 50
NSOCK INFO [0.0510s] nsock_write(): Write request for 27 bytes to IOD #1 EID 59 [172.21.100.133:53]
NSOCK INFO [0.0510s] nsock_write(): Write request for 27 bytes to IOD #1 EID 67 [172.21.100.133:53]
NSOCK INFO [0.0510s] nsock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 67 [172.21.100.133:53]
NSOCK INFO [0.0510s] nsock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 59 [172.21.100.133:53]
NSOCK INFO [0.1150s] nsock_trace_handler_callback(): Callback: READ SUCCESS for EID 34 [172.21.100.133:53] (27 bytes): Y.....target_ip.....
NSOCK INFO [0.1150s] nsock_read(): Read request from IOD #1 [172.21.100.133:53] (timeout: -1ms) EID 74
NSOCK INFO [0.1160s] nsock_trace_handler_callback(): Callback: READ SUCCESS for EID 74 [172.21.100.133:53] (27 bytes): Y.....target_ip.....
NSOCK INFO [0.1160s] nsock_read(): Read request from IOD #1 [172.21.100.133:53] (timeout: -1ms) EID 82
NSOCK INFO [0.5510s] nsock_ioc_delete(): nsock_ioc_delete (IOD #1)
NSOCK INFO [0.5510s] nevent_delete(): nevent_delete on event #82 (type READ)
NSOCK INFO [0.5510s] nsock_ioc_delete(): nsock_ioc_delete (IOD #2)
NSOCK INFO [0.5510s] nevent_delete(): nevent_delete on event #42 (type READ)
NSOCK INFO [0.5510s] nsock_ioc_delete(): nsock_ioc_delete (IOD #3)
NSOCK INFO [0.5510s] nevent_delete(): nevent_delete on event #50 (type READ)
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 5.85 seconds
^[[A^[[A^[[AStarting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:24 +0530
NSOCK INFO [0.0260s] nsock_ioc_new2(): nsock_ioc_new (IOD #1)
NSOCK INFO [0.0290s] nsock_connect_udp(): UDP connection requested to 172.21.100.133:53 (IOD #1) EID 8
NSOCK INFO [0.0290s] nsock_ioc_new2(): nsock_ioc_new (IOD #2)
NSOCK INFO [0.0290s] nsock_connect_udp(): UDP connection requested to 172.21.100.31:53 (IOD #2) EID 16
NSOCK INFO [0.0290s] nsock_ioc_new2(): nsock_ioc_new (IOD #3)
```